

ONLINE ACTIVITY #2  
Personalized Linux, and Git Cheat Sheet

## GIT & LINUX CHEAT SHEET

### THREE-TREE ARCHITECTURE

- **Working Dir:** Local sandbox (physical files checked out).
- **Staging Index:** Prep cache assembling changes for commit.
- **Repository (HEAD):** .git database. HEAD points to the current branch's latest commit.

### BASIC LINUX

#### **ls -lah**

List directory contents including hidden files.

#### **cd path/**

Change the current working directory.

#### **pwd**

Print current absolute path.

#### **mkdir -p dir**

Create new directories.

#### **rm -rf dir/**

Remove files or folders permanently.

#### **cp src dst**

Copy files or directories.

#### **mv src dst**

Move or rename files/folders.

#### **cat file**

Display complete file contents.

#### **grep "txt" f**

Search for text patterns inside files.

#### **chmod 755 f**

Change file access permissions.

### BASIC GIT

#### **git init**

Track new local repo.

Flags: --bare, -q

#### **git clone url**

Download existing remote repo.

Flags: --depth 1, --branch

#### **git status**

Show file tracking states.

Flags: -s

#### **git add .**

Stage changes for commit.

Flags: -p, -u

#### **git commit -m**

Record staged snapshot.

Flags: -a, --amend

#### **git restore f**

Discard local edits.

Flags: --staged

#### **git branch**

List/create branches.

Flags: -d, -a

#### **git switch**

Navigate branches.

Flags: -c (create & switch)

#### **git merge b**

Integrate branch history.

Flags: --no-ff, --abort

#### **git log**

View commit history.

Flags: --oneline, --graph

#### **git diff**

Show unstaged differences.

Flags: --staged

#### **git tag v1**

Attach permanent label.

Flags: -a, -d

### REFERENCES

- **Git Documentation:** [git-scm.com/doc](https://git-scm.com/doc)
- **Linux Man Pages:** [man7.org/linux/man-pages/](https://man7.org/linux/man-pages/)
- **Built-in Manual:** `git <command> --help`

*I have used gemini pro to do research for this work.*